

Where my skills at? A retrieval benchmark for skills toolkit

Anonymous Author(s)

Abstract

As the number of skills grows and agentic tasks run for longer, managing and selecting the right skills becomes increasingly difficult. This challenge is especially acute in dynamic repositories such as OpenClaw, where a new skill is added roughly every other day. In this work, we introduce a benchmark for evaluating how well retrieval systems can pull relevant skills from a skills repository under these changing conditions. To support this benchmark, we build a retrieval dataset with train and test splits and present a data generation process that adapts as the skill collection evolves over time.

We find that description-only retrieval fails almost completely in this setting, with all metrics below 1%, motivating richer summary-based representations. Using this setup, we compare strong dense embedding models from Qwen Embedding and a sparse BM25 baseline. Following our training recipe, we fine-tune a compact 0.6B embedding model that outperforms larger state-of-the-art models by an average of 30% across all retrieval metrics, with gains peaking at +38.7% on Hit@1. Notably, our trained model achieves strong dense-only performance, eliminating the need for a hybrid retrieval pipeline entirely.

Keywords: agent skills, retrieval benchmarks, dense retrieval, sparse retrieval, embedding models, software agents

1 Introduction

A skill is a reusable folder of instructions that helps an agent handle a task more accurately. Each skill is defined in a SKILL.md file with a name and a short description. The model uses these descriptions to decide which skills match a user request. Skills can be personal, so they are available across projects, or project-specific, so they are shared with a repository. They are loaded on demand when the system detects that they are relevant.

The main challenge is that selecting a skill often requires reading the name and description of many available skills from the file system before choosing the right one. As the skill collection grows, this process can waste context on many irrelevant skills and increase cost. We also observe that the description field, which is typically restricted to 1024 characters, is often insufficient to fully capture what a skill is capable of. In practice, critical operational details such as required environment variables, setup steps, dependencies, and debugging instructions are frequently omitted or only partially specified. As a result, important aspects of how to correctly invoke or troubleshoot a skill may be missing

or unclear. To address this, we introduce a compact summary representation of each skill that better reflects both its functional capabilities and operational requirements while remaining efficient for retrieval. Our goal is to move this filtering step out of the main model context and into a separate retrieval system. In this setup, the orchestrator model uses query-based tool calling: it sends the user request (or a rewritten version of it) to a retriever-skill tool, gets back the most relevant skill, and passes only that entire SKILL.md to the model.

Contributions.

- We release a retrieval dataset for agent skills that can help discover better strategies for skill selection.
- We evaluate current state-of-the-art embedding models, including Qwen3 Embedding[4], on this benchmark.
- We train a state-of-the-art embedding model for this retrieval setting.

2 Background and Related Work

Information retrieval (IR) is the task of finding relevant documents given a user query. In most systems, the query represents an information need, and the retriever returns a ranked list of documents that best match that need [1]. Classical approaches such as BM25 rely on exact term matching, while modern dense retrieval methods encode queries and documents into vectors and match them based on semantic similarity. Hybrid systems combine both approaches to balance efficiency and retrieval quality.

To evaluate these systems, the research community relies on standard benchmark datasets. One widely used benchmark is BEIR, which provides a collection of 18 datasets covering diverse retrieval tasks such as question answering, fact checking, and document search [2]. BEIR is designed to test how well retrieval models generalize across domains, and it has shown that simple methods like BM25 remain strong baselines, while dense models often require careful training to perform consistently well across tasks [2]. Other evaluation frameworks, such as MTEB, extend this idea by benchmarking embedding models across multiple tasks including retrieval, clustering, and classification [3].

However, these benchmarks assume a static corpus: the set of documents does not change during evaluation. In many real-world systems, this assumption does not hold. In addition, strong performance from dense retrievers often depends on task-related training data that adapts the model to the target retrieval setting. For example, in agent-based systems,

the retriever operates over a collection of skills or tools that are continuously updated. New skills are added, modified, or removed over time, which makes the retrieval problem more dynamic and increases the need for task-specific adaptation.

Recent work has started to study this setting explicitly. SkillFlow formulates skill selection as a retrieval pipeline over large skill collections [7]. SkillRouter shows that relying only on short metadata such as names and descriptions can significantly hurt performance, as important signals are often present in the full skill definition [8]. Similarly, Graph of Skills explores structured retrieval over dependencies between skills instead of treating them as independent documents [9].

In our setting, each document is a SKILL.md file describing a tool or capability, but its metadata is often incomplete: the description field is typically limited to around 1024 characters, so important details such as environment variables, dependencies, setup steps, and debugging instructions are often missing or only partially specified. This creates a gap between what the metadata says and what the skill can actually do, which makes short-description retrieval unreliable in a dynamic setting where both the skill collection and description quality keep changing.

3 Approach

3.1 Dataset Construction Pipeline

We construct the Dynamic Skills Benchmark through a four-stage pipeline designed to ensure data quality, diversity, and reproducibility. Figure 1 provides an overview of the full process.

Stage 1: Skill Sourcing. We source skills from the OpenClaw skills repository at a fixed snapshot (commit 7a7d8fe) to ensure full reproducibility. From this snapshot, we retain

only well-formed skills that contain both a name and a description field. Additionally, we discard any skill whose total context length exceeds 100K tokens, as excessively long contexts introduce noise and increase generation cost without a commensurate benefit to coverage.

Stage 2: Auxiliary Data Generation. For each retained skill, we use GPT-4o to generate three types of auxiliary data: (i) a concise natural-language summary of the skill, (ii) a set of seed questions grounded in the skill’s core functionality, and (iii) a set of persona-driven questions that simulate diverse user intents and interaction styles. The persona-driven questions are particularly important for capturing the breadth of real-world usage patterns that a retrieval system would encounter in deployment [10].

Stage 3: Question Filtering. We apply an automated filtering stage using Qwen2.5-72B Instruct to remove low-quality questions. Specifically, we discard questions that are (i) irrelevant to the associated skill, (ii) overly specific to implementation details rather than high-level intent, or (iii) insufficiently well-formed for evaluation purposes. This step substantially improves the clarity and generalizability of the resulting question pool.

Stage 4: Diversity-Aware Selection and Split. To reduce redundancy within each skill, we apply Maximal Marginal Relevance (MMR) [6] with Qwen3 Embedding 8B embeddings. MMR selects three diverse questions per skill. Two go to the training split and one goes to the test split, giving a clean separation between supervision and evaluation.

Model Training. We fine-tune a Qwen3-Embedding-0.6B model on the resulting training set. The model is trained with an InfoNCE objective [5], which encourages the embedding of a query to be closer to its associated skill than to other

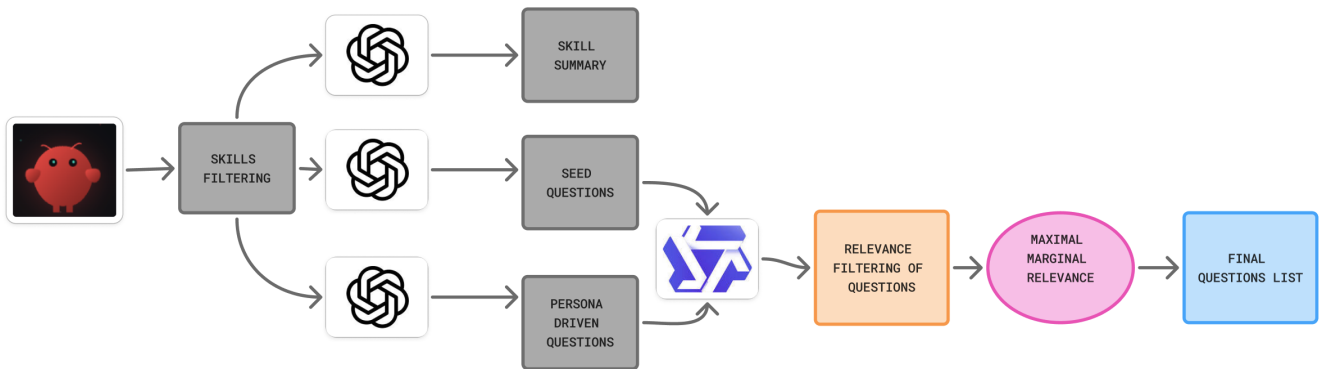


Figure 1. Dataset generation pipeline for the dynamic skills benchmark. We start from the OpenClaw skills GitHub repository at commit 7a7d8fe. We keep only properly formatted skills with both a name and description, and we remove skills whose context length exceeds 100K. We then use GPT-4o to generate skill summaries, seed questions, and persona-driven questions relevant to each skill. Next, we use Qwen2.5-72B Instruct to filter out irrelevant, overly skill-specific, or vague questions. Finally, we apply maximal marginal relevance to keep three diverse questions per skill: two for training and one for test.

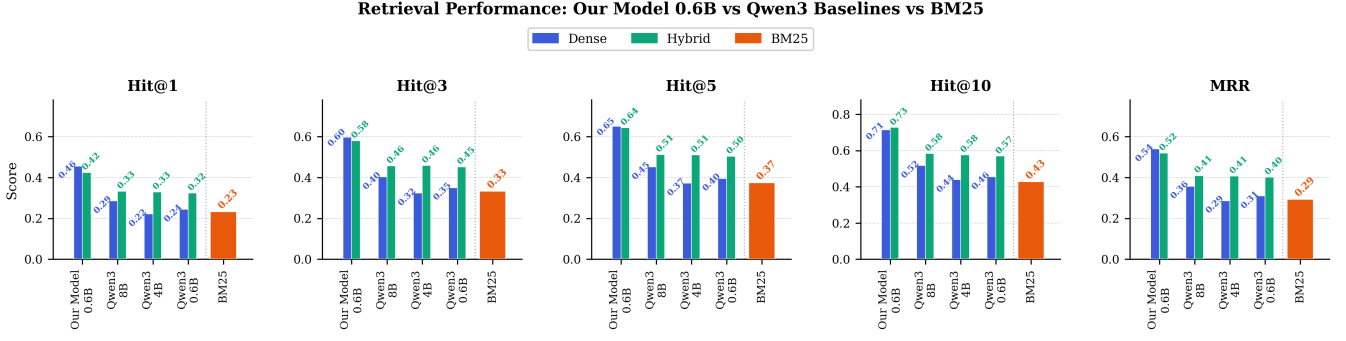


Figure 2. Retrieval performance comparison across Hit@K and MRR metrics.

skills in the batch. Despite its compact size (0.6B parameters, 2.4 GB in FP32), the trained model substantially outperforms larger un-trained baselines (see Section 4).

3.2 Method Overview

Before introducing our training pipeline, we outline the loss used for the embedding model during training. We use a contrastive loss based on the InfoNCE framework [5]. Given a batch of N training instances, the loss is defined as:

$$L_{\text{embedding}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s(q_i, d_i^+)/\tau}}{Z_i}, \quad (1)$$

where $s(\cdot, \cdot)$ is a similarity function, which we instantiate as cosine similarity, τ is a temperature parameter, and Z_i is the normalization factor that aggregates the similarity scores of the positive pair against multiple negative pairs:

$$Z_i = e^{s(q_i, d_i^+)/\tau} + \sum_k m_{ik} e^{s(q_i, d_{ik}^-)/\tau} + \sum_{j \neq i} m_{ij} e^{s(q_i, q_j)/\tau} + \sum_{j \neq i} m_{ij} e^{s(d_i^+, d_j^+)/\tau} + \sum_{j \neq i} m_{ij} e^{s(q_i, d_j^+)/\tau}. \quad (2)$$

Here, q_i is the query, d_i^+ is the positive document, and d_{ik}^- are negative documents. The function $s(\cdot, \cdot)$ denotes similarity (cosine), τ is a temperature parameter, and m_{ik}, m_{ij} are masking weights.

The term Z_i is the partition function that sums the exponentiated similarities of the positive pair with various negatives and in-batch pairs, ensuring the model ranks the positive pair higher than others.

3.3 Implementation Details

We train the Qwen3 Embedding 0.6B model for 1 epoch with a batch size of 128, a learning rate of 2.0×10^{-5} , and 200 warmup steps. We use hard negatives during training. For each query, we first retrieve relevant summaries and then keep 32 hard negatives after the top-5 results. We also treat summaries paired with other queries in the same batch as additional negatives for a given query. For hard-negative mining, we use a hybrid retriever that combines Qwen3

Embedding 8B with BM25, and we rank candidates using reciprocal rank fusion (RRF=60). In settings where we include a triplet loss term, we use a triplet margin of 0.2. Our final training set contains 60,249 unique queries, the held-out validation set contains 6,694 unique queries, and the test set contains 33,472 unique queries.

4 Evaluation

We evaluate retrieval performance across five standard metrics (Hit@1, Hit@3, Hit@5, Hit@10, and MRR) against Qwen3 baselines (0.6B, 4B, 8B) and BM25. **Description-only retrieval failed completely, with all metrics below 1% showing that descriptions are insufficient.** As shown in Figure 2, our trained 0.6B model substantially outperforms all baselines across every metric. Notably, the dense retrieval variant of our model matches or exceeds its hybrid counterpart (e.g., Hit@1: 0.46 dense vs. 0.42 hybrid; MRR: 0.54 vs. 0.52), demonstrating that fine-tuning alone is sufficient to close the gap typically bridged by augmenting dense retrieval with sparse BM25 signals. This is a practically significant

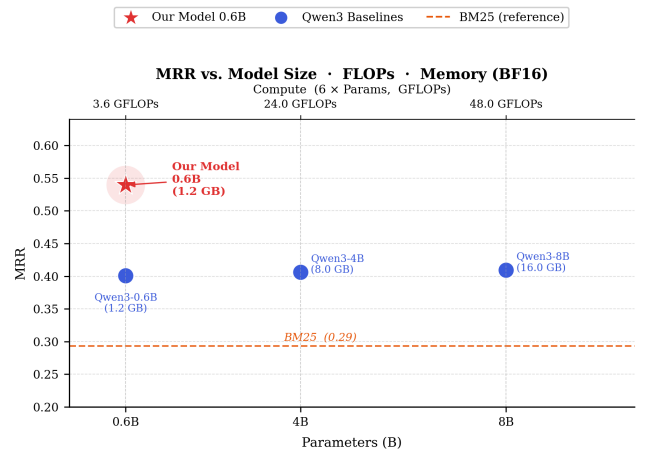


Figure 3. MRR-efficiency comparison for the evaluated retrieval systems vs. the compute and memory footprint

result: by achieving strong dense-only performance, we eliminate the need for a hybrid pipeline, removing the computational overhead of maintaining and querying a BM25 index at inference time. In contrast, the un-trained Qwen3 models consistently benefit from hybrid search over dense alone, underscoring that the gains from our training procedure render sparse retrieval redundant.

Despite being 13× and 6× smaller than Qwen3-8B and Qwen3-4B respectively, our model outperforms both by an average of 30% across all metrics. The gains are most pronounced at Hit@1, where our model surpasses Qwen3-8B by +37.3% and Qwen3-4B by +38.7%, and remain substantial at MRR (+31.8% and +32.8%), which directly reflects ranking quality. This demonstrates that targeted training yields a compact model that simultaneously reduces inference cost and outperforms much larger un-trained alternatives, making it well-suited for production retrieval systems where both latency and accuracy are critical.

In the pareto-efficiency graph (Figure 3), our model achieves the best performance-to-computational-footprint ratio, further validating our training methodology.

5 Discussion and Limitations

Our results show that description-only retrieval is not a viable strategy for skill selection in dynamic agent settings. In our benchmark, it fails almost completely, which supports the intuition that short metadata fields do not capture the operational details that often determine whether a skill is actually relevant. This finding motivates richer intermediate representations, such as summary-based retrieval, that preserve more of a skill’s functionality and execution requirements while remaining compact enough for efficient indexing.

The results also suggest that diversity in query generation is important for training robust retrievers. Seed questions provide direct coverage of a skill’s core functionality, while persona-driven questions broaden the distribution of intents, phrasings, and user needs associated with the same underlying skill. Together, these complementary question types improve the diversity of the supervision signal and likely contribute to better generalization at evaluation time.

Limitations. Although we evaluate on OpenClaw, to our knowledge the largest openly available skills repository, the study still does not fully capture the systems challenges that emerge in a truly large-scale skills data hub. As the number of skills grows substantially further, indexing, retrieval efficiency, and update management may become considerably more difficult. In addition, our benchmark assumes a fixed snapshot of the skill repository during dataset construction. In real deployments, skills change over time; when they are updated dynamically, faithful summaries may also need to be regenerated so that the retrieval index stays aligned with the current behavior and requirements of each skill.

6 Conclusion

We introduced a dynamic retrieval benchmark for agent skills that better reflects deployment settings where skill collections evolve over time. Our results show that short description-based retrieval is insufficient in this setting, while richer summary-based representations provide a stronger foundation for retrieval. We also find that combining seed questions with persona-driven questions produces more diverse supervision and helps train a compact embedding model that outperforms substantially larger un-trained baselines. Together, these findings suggest that skill retrieval should be treated as a dynamic, representation-sensitive problem, and that targeted training on realistic query distributions is a promising path toward robust retrieval for agent systems.

References

- [1] C. J. van Rijsbergen. 1979. Information Retrieval. Butterworth-Heinemann.
- [2] Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models. *NeurIPS Datasets and Benchmarks*.
- [3] Muennighoff et al. 2023. MTEB: Massive Text Embedding Benchmark. *arXiv preprint arXiv:2210.07316*.
- [4] Yanzhao Zhang et al. 2025. Qwen3 Embedding: Advancing Text Embedding and Reranking through Foundation Models. *arXiv preprint arXiv:2506.05176*.
- [5] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. 2018. Representation Learning with Contrastive Predictive Coding. *arXiv preprint arXiv:1807.03748*.
- [6] Jaime Carbonell and Jade Goldstein. 1998. The Use of MMR, Diversity-Based Reranking for Reordering Documents and Producing Summaries. In *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*.
- [7] Pagkratios Tagkopoulos, Fangzhou Li, and Ilias Tagkopoulos. 2025. SkillFlow: Efficient Skill and Code Transfer Through Communication in Adapting AI Agents. *arXiv preprint arXiv:2504.06188*.
- [8] YanZhao Zheng et al. 2026. SkillRouter: Retrieve-and-Rerank Skill Selection for LLM Agents at Scale. *arXiv preprint arXiv:2603.22455*.
- [9] Dawei Li et al. 2026. Graph of Skills: Dependency-Aware Structural Retrieval for Massive Agent Skills. *arXiv preprint arXiv:2604.05333*.
- [10] Tao Ge et al. 2024. Scaling Synthetic Data Creation with 1,000,000,000 Personas. *arXiv preprint arXiv:2406.20094*.